X-LINKED ICHTHYOSIS

The scales are large and dark and most evident on the flexural areas in this patient. The opacities appear white on slit-lamp examination of the cornea.

Serum cholesterol sulfate levels are typically elevated in X-linked ichthyosis (2000 to 9000 μg/dL), compared to the normal range of 80 to 200 μg/dL. Unfortunately, cholesterol sulfate determination in serum and skin scales may not be readily available for laboratory confirmation. In rare cases where deletions are not detected but clinical suspicion remains high, arylsulfatase C (steroid sulfatase) enzyme activity can be measured in cultured fibroblasts.

Deletions involving adjacent sulfatase genes can explain overlap syndromes that include chondrodysplasia punctata and X-linked ichthyosis. Additionally, X-linked Kallmann syndrome—characterized by hypogonadotropic hypogonadism, anosmia, and sometimes renal abnormalities, obesity, synkinesis, cleft palate, and spastic paraplegia—can co-occur with X-linked recessive ichthyosis as part of a contiguous gene deletion syndrome.

Because of this association and the increased risk of testicular carcinoma, patients with X-linked recessive ichthyosis should be questioned about anosmia and undergo periodic testicular examinations.

In the epidermis, steroid sulfatase catalyzes the hydrolysis of cholesterol sulfate. The deficiency of this enzyme in X-linked ichthyosis underscores the importance of cholesterol sulfate hydrolysis in normal desquamation. Topical application of cholesterol sulfate in mice induces a scaling disorder, further supporting its role in corneocyte shedding.

A family has been described with an X-linked inheritance pattern for ichthyosis but with normal steroid sulfatase activity and absence of corneal opacities. This finding demonstrates genetic heterogeneity within X-linked ichthyosis. Although steroid sulfatase deficiency accounts for the majority of cases, a normal enzyme level in a male with ichthyosis does not exclude X-linked inheritance.

AUTOSOMAL RECESSIVE CONGENITAL ICHTHYOSIS (ARCI)

Autosomal recessive congenital ichthyosis (ARCI) is a heterogeneous group of disorders presenting at birth with generalized skin involvement. ARCI may be syndromic (associated with extracutaneous manifestations) or nonsyndromic. It is rare, with an estimated prevalence of 1 in 300,000 individuals.

Historically, non-bullous congenital ichthyosiform erythroderma (NCIE) and lamellar ichthyosis (both autosomal recessive) were distinguished from bullous congenital ichthyosiform erythroderma (epidermolytic hyperkeratosis, autosomal dominant) based on clinical features such as the presence of bullae and inheritance pattern. However, the interchangeable use of "lamellar ichthyosis" with NCIE—and the inclusion of a spectrum of phenotypes under this term—has led to diagnostic confusion.

Williams and Elias differentiated lamellar ichthyosis from congenital ichthyosiform erythroderma (CIE), the latter being a milder, erythrodermic form. Clinically, lamellar ichthyosis is characterized by large, dark, plate-like scales. Although affected infants may be erythrodermic at birth, adults typically show less redness and more prominent scaling compared to those with CIE.